

AKASH S

AI/ML and Software Engineer who builds and ships RAG systems, agentic workflows, fine-tuned speech models, voice-first AI, and developer tools. Strong backend engineering, clean product thinking.

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Visvesvaraya Technological University

Bengaluru, Karnataka

Bachelor of Engineering in Information Science and Technology | CGPA: 8.38 / 10 Aug 2021 – Jun 2025

Courses: Data Structures and Algorithms, Database Management Systems, Operating Systems, Machine Learning, Computer Networks

SKILLS SUMMARY

- **Languages:** Python, TypeScript, JavaScript, SQL
- **AI / GenAI:** Generative AI, LLMs, RAG, Prompt Engineering, AI Agents, Agentic Workflows, LangChain, LangGraph, Vector Databases
- **AI Dev Tools:** Cursor, Claude Code, OpenAI Codex, LangSmith, Weights & Biases, GitHub Copilot
- **ML / Fine-Tuning:** Model Fine-Tuning, LoRA, HuggingFace, Transformers, Whisper, Scikit-learn, Pandas, NumPy
- **Frameworks:** FastAPI, React.js, Next.js, Node.js
- **Cloud & DevOps:** Microsoft Azure, Docker, CI/CD Pipelines, Vercel
- **Databases:** PostgreSQL, ChromaDB, SQLAlchemy

EXPERIENCE

BGS Infotech

Bengaluru, Karnataka

AI Software Intern

Sep 2025 – Apr 2026

- **Medical AI Research:** Developed a liver tumor classification model using image processing and ML techniques, findings contributed to a co-authored research paper on medical AI diagnostics.
- **Client Feature Delivery:** Built and delivered e-commerce platform features across the full development lifecycle, handling real client requirements from specification through production deployment.
- **AI Tool Adoption:** Led internal adoption of AI-powered developer tooling, improving engineering productivity and code review practices across the team.

Prime Minds Consultancy

Bengaluru, Karnataka

AI Intern

Feb 2025 – May 2025

- **GenAI NL Query Interface:** Engineered a Generative AI-powered natural language query interface for Odoo ERP using RAG pipelines and LLM orchestration, enabling non-technical users to retrieve enterprise data in plain English.
- **RAG Pipeline Development:** Designed retrieval-augmented generation pipelines integrating PostgreSQL with vector search for grounded, contextual LLM responses across business datasets.
- **Workflow Automation:** Automated data extraction and validation workflows in Python, reducing manual processing effort and improving operational pipeline reliability.

Kennametal Inc.

Bengaluru, Karnataka

Project Intern

May 2024 – Nov 2024

- **Computer Vision QC:** Developed a computer vision quality control system for manufacturing defect detection using image processing techniques on production-grade industrial datasets.
- **Image Processing Pipeline:** Built 2D image extraction and preprocessing pipelines supporting automated defect analysis and reducing manual inspection in manufacturing workflows.

PROJECTS

- **Nexa RAG AI: Enterprise Document Intelligence Platform** [\[Link\]](#)
 - Architected an end-to-end RAG platform for PDF, DOCX, and TXT ingestion using MiniLM embeddings, ChromaDB vector search, top-8 retrieval, confidence scoring, and citation-backed LLM response.
 - Built LangGraph workflows for multi-document summarization, comparison, action-item extraction, and professional email draft generation from uploaded documents.
 - Engineered FastAPI backend with JWT authentication, SQLAlchemy ORM, PostgreSQL persistence, 15+ REST endpoints, and pluggable LLM providers including Groq, OpenAI, Anthropic, and Gemini.
 - Containerized the system with Docker Compose across API, PostgreSQL, ChromaDB, and React frontend services using environment-based configuration and persistent volumes..
- **ForKrishi AI: Voice-First Agricultural Intelligence Platform** [\[Link\]](#)
 - Built a multilingual voice-first crop disease diagnosis system for Indian farmers using Gemini vision API for leaf analysis with bounding-box overlays, STT/TTS in Kannada, Hindi, and English.
 - Implemented an offline-capable PWA with service worker caching, geolocation-based KVK discovery, and voice-enabled Mandi price search for field usage.
 - Developed canvas-based advisory PNG export for WhatsApp sharing and deployed the application on Vercel using Next.js serverless routes.
- **Hindi ASR Fine-Tuning: Whisper Small with LoRA** [\[Link\]](#)
 - Fine-tuned OpenAI Whisper Small for Hindi speech recognition using LoRA, training only ~0.8% of parameters on Google FLEURS Hindi data with Kaggle T4 GPU.
 - Achieved **49% relative WER reduction** from 61.0% to 31.2% and **63% relative CER reduction** from 30.7% to 11.2% on a 418-sample held-out test set.
Benchmarked Whisper Tiny, Base, Small, and Medium; designed a FastAPI /transcribe deployment roadmap with JWT auth, rate limiting, and ONNX optimization
- **Druva AI: Streaming Developer Chat Assistant** [\[Link\]](#)
 - Built a real-time AI developer chat assistant with Groq-powered SSE streaming, async generators, and progressive React state updates for fast conversational UX.
 - Implemented voice-to-text input, localStorage-based chat history, math-formula rendering, IST-aware prompt context, and responsive light/dark/OLED themes.
 - Designed it as an AI assistant prototype focused on developer productivity, fast responses, and hands-free interaction.

ACHIEVEMENTS

- **Winner, Zero Boundaries Hackathon:** Built an AI-powered multiplayer coding challenge platform with automated code review, real-time debugging suggestions, and competitive room-based gameplay.
- **Co-Author, Research Paper on Liver Tumor Classification:** Applied ML and image processing to medical imaging datasets; findings co-authored and submitted for publication.
- **Cricket Champion:** Led team to championship at SKIT Premier League, demonstrating team leadership under competitive conditions.

CERTIFICATIONS

- Generative AI Engineering and Fine-Tuning Transformers, IBM, 2026
- Fundamentals of AI Agents Using RAG and LangChain, IBM, 2026
- Microsoft Azure Cloud Services, Microsoft, 2025